

Facilitator effects in Mandarin topicalization: Evidence for a crossing-based view of anti-locality*

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1. Introduction

A major inquiry in Minimalist syntax concerns how movement is constrained. In particular, *anti-locality* constraints rule out movement that is too local (see e.g., Bošković 1994, Saito and Murasugi 1999, Abels 2003, Grohmann 2003, Erlewine 2016, 2020, Newman 2020, Zyman 2021, Branen 2022, a.o.). Much of the literature defines anti-locality as a requirement of crossing which also depends on specific *structural positions* from which and to which movement occurs. Specifically, *Comp-to-Spec anti-locality* requires that movement of a phrase from Comp, X must cross a head other than X (see e.g., Abels 2003), thereby ruling out movement from Comp to Spec of the same projection, but allowing movement from Comp, X to Spec, YP, crossing Y, as illustrated in (1a). *Spec-to-Spec anti-locality* requires that movement of a phrase from Spec, XP must cross a maximal projection other than XP (see e.g., Erlewine 2016, 2020, Bošković 2016, Deal 2019, Branen 2022), thereby ruling out movement from Spec, XP to Spec, YP, but allowing movement from Spec, XP to Spec, ZP, crossing YP (and Z), as illustrated in (1b).¹

Recently, Newman (2020) proposes a unified *crossing*-based view of anti-locality, without resorting to specific positions. Informally, in both (1a) and (1b), the movement is licit if and only if the moved phrase crosses an *additional* constituent on the movement path, i.e., Y in (1a) and Z in (1b), which is distinct from the closest head—X in (1a) and Y in (1b)—that c-commands it. Newman motivates this crossing-based view of anti-locality through *facilitator effects* observed in middles across languages, where a *high facilitator* (above vP, e.g., negation) or a *low facilitator* (within vP, e.g., an adverb) is required for the construction to be well-formed (e.g., *The bread didn't cut. The bread cut easily.*).

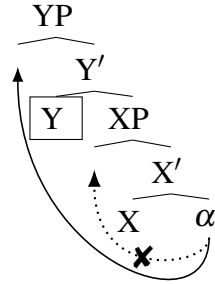
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¹For other types of anti-locality constraints, see Grohmann (2011), Richards (2022) for overviews.

(1) *Position-based vs. Crossing-based anti-locality*

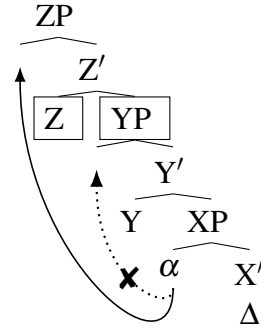
a. *Comp-to-Spec anti-locality*

- ✗ Comp, X to Spec, XP movement
- ✓ Comp, X to Spec, YP movement, crossing Y.



b. *Spec-to-Spec anti-locality*

- ✗ Spec, XP to Spec, YP movement
- ✓ Spec, XP to Spec, ZP movement, crossing YP (and Z).



In this paper, we demonstrate that topicalization in Mandarin exhibits a similar and peculiar kind of facilitator effects, requiring *either* a high facilitator, such as a temporal adverb, *or* a low facilitator, such as a manner adverb, as seen in (2).

(2) a. *(IP-)internal topicalization in non-finite clauses: banned without facilitators*

*Lisi bipo wo_i [PRO_i **zhe-jian shi** zuo _].

Lisi force 1SG this-CL matter do

Int. 'Lisi forced me to, this matter, do (it).'

b. *... amelioratable by crossing a high facilitator (e.g., temporal adverb)*

Lisi bipo wo_i [PRO_i **zhe-jian shi** mingtian zuo _].

Lisi force 1SG this-CL matter tomorrow do

Lit. 'Lisi forced me to, this matter, do (it) tomorrow.'

c. *... amelioratable by crossing a low facilitator (e.g., manner adverb)*

Lisi bipo wo_i [PRO_i **zhe-jian shi** manman/toutou zuo _].

Lisi force 1SG this-CL matter slowly/secretly do

Lit. 'Lisi forced me, this matter, slowly/secretly do (it).'

We argue that such facilitator effects support a unified crossing-based view of anti-locality, conditioned by the *Principle of Minimal Compliance* (PMC, Richards 1998). Specifically, we argue that topicalization in Mandarin involves *two* movement steps that each may violate a different kind of position-based anti-locality constraints—Comp-to-Spec and Spec-to-Spec—but, crucially, only requires *one* facilitator for *either* movement step. We propose that such facilitator effects arise from the combination of the crossing-based view of anti-locality (Newman 2020), which unifies Comp-to-Spec and Spec-to-Spec anti-locality constraints, and the PMC, which allows anti-locality to be *evaluated only once* for a successive-cyclic movement chain (cf. Toquero-Pérez 2022 for a similar idea of applying PMC to anti-locality, in a different empirical domain and with a very different implementation).

The remainder of the paper is organized as follows: Section 2 provides a primer on Mandarin topicalization; Section 3 examines facilitator effects in Mandarin topicalization; Section 4 presents an anti-locality account; and Section 5 concludes.

2. A primer on Mandarin topicalization

In Mandarin, topicalization can target an IP-internal or IP-external position, either below or above the subject in Spec, IP, as seen in (3) (see e.g., Qu 1994, Shyu 1995, Ting 1995, Paul 2002, 2005, Kuo 2009, Chen 2023, 2025). In this section, we present three pieces of evidence to establish that (IP-)internal and (IP-)external topicalization involve successive-cyclic movement—via Spec, $vP(s)$ —into the low IP area (Spec, $TopP_{Internal}$) and the left periphery (Spec, $TopP_{External}$), respectively, as schematized in (4).

- (3) *(IP-)external vs. (IP-)internal topicalization*
 {**Zhe-jian shi_i**} [_{IP} Lisi {**zhe-jian shi_i**} zuo-le _i].
 this-CL matter Lisi this-CL matter do-PFV
 ‘{This matter}, Lisi, {this matter}, did (it).’
- (4) a. *Step I: Intermediate movement to Spec, vP*
 ... [_{TopP} [_{TopP'} Top ... [_{vP} **Obj_i** [_{v'} $v+V$ [_{VP} t_V _i ...
 b. *Step II: Terminating movement to Spec, $TopP_{Internal/External}$*
 ... [_{TopP} **Obj_i** [_{TopP'} Top ... [_{vP} _i [_{v'} $v+V$ [_{VP} t_V _i ...

First, it is well established that both internal and external topicalization involve movement, as evidenced by their *sensitivity to (complex NP) islands*, as seen in (5).²

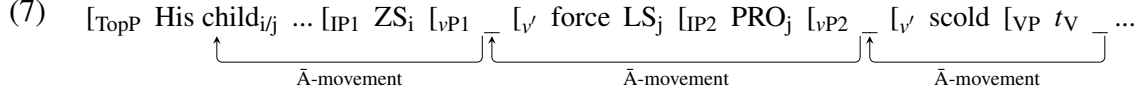
- (5) {***Zhe-ge ren_i**} LS {***zhe-ge ren_i**} zhuazou-le [_{NP} na-ge sha-le _i de ren].
 this-CL person LS this-CL person arrest-PFV that-CL kill-PFV REL person
 ‘{*This person}, Lisi, {*this person}, arrested that person who killed (him).’

Second, Chen (2023, 2025) has argued that both internal and external topicalization involves successive-cyclic movement via Spec, $vP(s)$. In (6), external topicalization involves $\bar{A}$ -movement and shows *Principle A reconstruction effects* (see also Qu 1994, Ting 1995). In contrast, internal topicalization involves composite A/ $\bar{A}$ -movement and hence does not show Principle A reconstruction effects.

- (6) a. Zhangsan_i bi Lisi_j ma-guo **ta-ziji_j/*i-de haizi**.
 Zhangsan force Lisi scold-EXP 3SG-self's child
 ‘Zhangsan_i forces Lisi_j to scold his_j/*i child.’ (baseline)

²See Huang et al. (2009) for additional tests of movement; see also Huang (1982, 1987), Li (1990, 2000), Qu (1994), Ting (1995), Shyu (1995), Kuo (2009), Chen (2023, 2025) a.o.

- b. {**Ta-ziji_{i/j}-de haizi**}, ZS_i {**ta-ziji_{i/*j}-de haizi**} bi LS_j ma-guo ____.
 3SG-self's child ZS 3SG-self's child force LS scold-EXP
 ‘{His_{i/j} child}, ZS_i, {his_{i/*j} child}, forces LS_j to scold (him).’ (*topicalization*)



Nevertheless, Chen (2023, 2025) has also argued, based on the lack of reconstruction in certain configurations, that external topicalization may alternatively involve intermediate step(s) of composite A/ $\bar{A}$ -movement to Spec, vP(s). This supports the view that the Mandarin *v* hosts not only a pure $\bar{A}$ -probe but also a composite A/ $\bar{A}$ -probe for purposes of successive-cyclic movement. Following Chen (2023, 2025), we take this as indirect evidence that internal topicalization also proceeds successive-cyclically via Spec, vP(s).

Third, we assume that topicalization targets the Spec of a TopP, which may be either internal or external to IP (see e.g., Paul 2002, 2005, Chen 2023, 2025). While the external TopP is projected in the left periphery, the internal TopP has variable projection sites within the low IP area (cf. Kuo 2009). While an internal topic may occur either above or below a deontic modal (see Tsai 2015 on the IP-internal position of ModP^{Deo}), it must surface above and not below an aspectual projection (AspP) when it is present (which rules out a vP adjunction analysis of internal topicalization, see section 4.2), as illustrated in (8–9).

- (8) a. [IP Lisi {**zhe-jian shi_i**} yinggai {**zhe-jian shi_i**} haohao zuo i].
 Lisi this-CL matter should this-CL matter well do
 Lit. ‘Lisi, this matter, should do (it) well.’ (*deontic modals*)
- b. [IP Lisi {**zhe-jian shi_i**} (zheng)zai {***zhe-jian shi_i**} haohao zuo i].
 Lisi this-CL matter PROG this-CL matter well do
 Lit. ‘Lisi, this matter, is investigating (it) well.’ (*progressive aspect*)
- c. [IP Lisi {**zhe-jian shi_i**} hai meiyou {***zhe-jian shi_i**} haohao zuo i].
 Lisi this-CL matter yet NEG.PFV this-CL matter well do
 Lit. ‘Lisi, this matter, hasn’t yet done (it) well.’ (*aspectual negation*)

- (9) [CP Δ_i ... [IP Subj ... Δ_i ... [ModP^{Deo} Δ_i ... [AspP * Δ_i ... [vP * Δ_i ... [VP V i ...
- Possible sites of TopP*

3. Facilitator effects in Mandarin topicalization

It has long been observed that internal topicalization is unavailable within control verb complements, as seen in (10). This restriction has led many to treat the (un)availability of internal topicalization as a diagnostic of finiteness (see e.g., Huang 1982, 1989, Li 1990, Tang 2000, Lin 2011, 2012, 2015, i.a.) or the presence (or lack) of a full IP/TP layer in the non-finite complement (see e.g., Grano 2012, 2015, N. Huang 2018, C.-T. J. Huang 2022).

- (10) a. *Lisi_i dasuan/changshi/keyi [PRO_i **zhe-jian shi** zuo _].
 Lisi plan/try/be.allowed this-CL matter do
 Int. ‘Lisi plans/tries/is allowed to, this matter, do (it).’ (Subj. control)
- b. *Lisi bi(po)/mingling/quan wo_i [PRO_i **zhe-men ke** xuan _].
 Lisi force/order/urge 1SG this-CL class choose
 Int. ‘Lisi forced/ordered/urged me to, this course, take (it).’ (Obj. control)

In this section, we demonstrate a new generalization: the ill-formedness of (10) can be ameliorated by the presence of an additional element, a *facilitator*, on the topicalization path—specifically, a *high facilitator* located between Spec, vP and Spec, TopP, or a *low facilitator* located between Comp, V and Spec, vP; more generally, we argue that internal topicalization always requires either a high or a low facilitator, while external topicalization always has a high facilitator, the subject in Spec, IP. We contend that internal topicalization is *not* a test of finiteness, nor IP/TP clause size.

3.1 High and low facilitators

N. Huang (2018) and C.-T. J. Huang (2022), among others, have observed that internal topicalization within a non-finite clause becomes available with the presence of a high facilitator located between Spec, vP and Spec, TopP, such as a temporal adverb in (11a) or negation in (11b). On the basis of such data, they argue that internal topicalization requires a clause of IP/TP size, as opposed to a smaller size, e.g., vP.

- (11) *Internal topicalization in non-finite clauses* (Huang 2022:ex.77–78, adapted)
- a. ... *amelioratable by crossing a high facilitator* (e.g., temporal adverb)
 Lisi dasuan/ changshi/ bi wo [PRO **zhe-men ke** ming-nian xuan _].
 Lisi plan try force 1SG this-CL class next-year choose
 Lit. ‘Lisi plans/tries/forces me to, this class, next year take (it).’
- b. ... *amelioratable by crossing a high facilitator* (e.g., negation)
 Lisi dasuan/ changshi/ bi wo [PRO **zhe-men ke** bu xuan _].
 Lisi plan try force 1SG this-CL class NEG choose
 Lit. ‘Lisi plans/tries/forces me to, this class, not take (it).’

We further observe that internal topicalization within non-finite clauses also becomes available with the presence of a *low* facilitator located between Comp, V and Spec, vP, such as a pre-verbal manner adverb (12a) or a post-verbal degree modifier headed by *-de* (DegreeP) (12b) (see section 3.2 for more details).³ While we have independently shown in section 2.3 that internal topicalization lands above AspP (if exists), note that the presence of a low facilitator is not contingent on the presence of an IP/TP layer. We thus contend that inter-

³The in-situ counterpart of (12b) requires so-called *verb-copying*—that is, the appearance of two instances of the verb: one preceding the object and another preceding the DegreeP—resulting in a V_i-Obj-V_i-DegreeP configuration (see e.g., Huang 1982, Gouguet 2006, Cheng 2007). While verb-copying has been a topic of much discussion, it is not the focus of this paper.

nal topicalization does *not* require an IP/TP layer (*pace* Huang 2022 a.o.). Also note that facilitators need *not* linearly precede the launching site of the object, as seen in (12b).

(12) *Internal topicalization in non-finite clauses*

- a. ... *amelioratable by crossing a low facilitator* (e.g., *manner adverb*)
 Lisi dasuan/changshi/bi wo [PRO **zhe-jian shi** manman/toutou zuo _].
 Lisi plan/try/force 1SG this-CL matter slowly/secretly do
 ‘Lisi plans/tries/forces me to, this matter, slowly/secretly do (it).’
- b. ... *amelioratable by crossing a low facilitator* (e.g., *degree modifier*)
 Lisi dasuan/ changshi/ bi wo [PRO **kafei** [_V he-de
 Lisi plan try force 1SG coffee drink-DE
 [_{VP} [_V he _] [_{DegreeP} de kuai yi-dian]]].
 fast a-bit
 ‘Lisi plans/tries/forces me, coffee, drink (it) a bit faster.’

Before proceeding, a few words are in order about external topicalization. First, unlike internal topicalization, which always requires either a high or a low facilitator, external topicalization does not require any additional facilitators. We suggest this is because external topicalization always has a high facilitator—the subject in Spec, IP. Second, we assume that external topicalization is unavailable within non-finite clauses due to the lack of a left periphery; thus, we have consistently treated a topicalized object within a non-finite clause as occupying an IP-internal position, below the PRO subject.

3.2 Non-facilitators

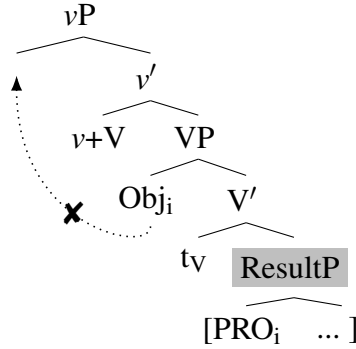
Importantly, internal topicalization cannot be facilitated by just *any* additional element—any element located outside the movement path is *not* a facilitator. For example, in (13), the post-verbal resultative phrase (ResultP) is arguably located in Comp, V, structurally lower than the object’s launching site (Spec, VP, in this case) (see e.g., Chen 2023). As a result, it fails to facilitate topicalization. An additional facilitator is required, such as an adverb (e.g., *guyi* ‘intentionally’).

(13) *Internal topicalization in non-finite clauses*

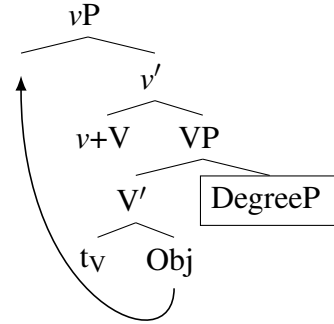
- ... *not amelioratable by a non-facilitator* (e.g., *resultative phrase*)
 Lisi dasuan/ changshi/ bi wo [PRO **na-ge haizi** [_V *(guyi) ma-de
 Lisi plan try force 1SG that-CL child intentionally scold-DE
 [_{VP} _ [_V ma [_{ResultP} de PRO_i da ku]]]].
 loudly cry
 ‘Lisi plans/tries/forces me to, that child, *(intentionally) scold (him) as a result of which (he) would cry loudly.’

ResultP contrasts with DegreeP, as seen in (12b). We suggest that DegreeP is a VP-level adjunct, structurally higher than the object’s launching site, hence facilitates topicalization.

(14) a. *ResultP as non-facilitator*



b. *DegreeP as low facilitator*



3.3 Facilitator effects in finite clauses

The same facilitator effects carry over to (generic) finite clauses without an AspP, where we also observe a contrast between internal and external topicalization, as in (15) (as opposed to the lack of such a contrast in (3), where *-le* indicates the presence of an AspP).⁴

- (15) *Context: Who drinks coffee (habitually)?*
 { **Kafei_i** } [_{IP} Lisi { ***kafei_i** } he _i].
 coffee Lisi coffee drink
 ‘{Coffee}, Lisi, { *coffee }, drinks (it).’

Within (generic) finite clauses, internal topicalization also requires the presence of a high facilitator in (16a), or a low facilitator in (16b). By contrast, external topicalization is licit without any of the aforementioned facilitators, as seen in (15), which, as we have suggested, is because the subject in Spec, IP serves as a high facilitator.

(16) *Internal topicalization in finite clauses*

- a. ... *amelioratable by crossing a high facilitator (e.g., temporal adv./negation)*
 Lisi **kafei_i** mei-tian/bu he _i.
 Lisi coffee every-day/NEG drink
 ‘Lisi, coffee, every day/does not drink(s) (it).’
- b. ... *amelioratable by crossing a low facilitator (e.g., VP modifier)*
 Lisi **kafei_i** he-de _i [_{DegreeP} **de** hen kuai].
 Lisi coffee drink-DE very fast
 ‘Lisi, coffee, drinks (it) fast.’

⁴We assume that the generic operator Gen attaches high, at the IP/TP level rather than AspP. Evidence comes from subjects (at Spec, IP) with generic readings, which require licensing by a higher Gen (e.g., impersonal pronouns; see Chierchia 1995, Soares 2021, and Tsai 2001 for Chinese).

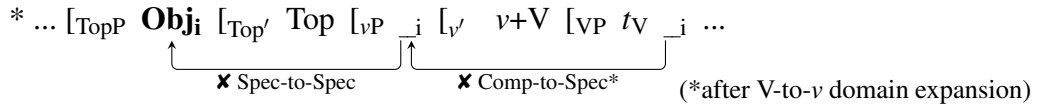
4. An anti-locality account

In this section, we propose that the ungrammaticality of topicalization without facilitators in Mandarin is a result of *anti-locality* violations. Specifically, we propose in section 4.1 that the facilitator effects arise from the combination of two components in the grammar: the unified *crossing*-based anti-locality (Newman 2020) and the *Principle of Minimal Compliance* (Richards 1998). We also present arguments against alternative analyses in section 4.2–4.3.

4.1 The requirement of crossing and the Principle of Minimal Compliance

As previously discussed, topicalization in Mandarin involves intermediate movement step(s) to Spec, ν P and a terminating movement step to a Spec, TopP either internal or external to IP (see (4)). In cases where the topicalized phrase originates from Comp, V, the intermediate movement step to Spec, ν P would incur a Comp-to-Spec anti-locality violation (with an additional assumption that V-to- ν head movement expands the locality domain, see (19) below). Moreover, in cases where the topicalized phrase targets Spec, TopP_{Internal}, the terminating movement step would incur a Spec-to-Spec anti-locality violation.

(17) *Anti-locality violations in internal topicalization*



However, as seen in section 3.1, only *one* facilitator—either a high facilitator located between Spec, ν P and Spec, TopP or a low facilitator located between Comp, V and Spec, ν P—is required in order to circumvent the *two* different kinds of position-based anti-locality violations. This gives rise to a puzzle: how can a low facilitator, which ameliorate a Comp-to-Spec anti-locality violation incurred on the intermediate movement step, *also* ameliorate a Spec-to-Spec anti-locality violation incurred on the terminating movement step (and vice versa for a high facilitator)?

We propose to account for the facilitator effects in Mandarin topicalization with the following two components. First, we adopt Newman’s (2020) crossing-based view of anti-locality in (18), which unifies Comp-to-Spec and Spec-to-Spec anti-locality constraints. We also assume with Newman (2020) that head movement expands the locality domain (19), such that movement from Comp, V to Spec, ν P without a low facilitator would incur an anti-locality violation due to V-to- ν head movement.

(18) *Crossing-based anti-locality* (Newman 2020:6)

- a. *Anti-locality*: A moving constituent α must cross a constituent distinct from the closest X^0 that c-commands it.
- b. *Crossing*: Movement from a position α to a position β crosses a constituent γ iff γ asymmetrically c-commands α but not β .

- (19) *Head movement expands locality domain* (Newman 2020:21, adapted)
 If X^0 moves to Y^0 , and X^0 was the closest element that c-commanded α , then Y^0 is no longer distinct from X^0 in the evaluation of (18).

Second, we propose that the PMC in (20), which, informally, suspends a constraint (for a given Goal) as long as it is satisfied once (with the given Goal), applies to anti-locality constraints, allowing it to be evaluated only once for a successive-cyclic movement chain.⁵

- (20) *Principle of Minimal Compliance* (Richards 1998:601)
For any dependency D that obeys constraint C, any elements that are relevant for determining whether D obeys C can be ignored for the rest of the derivation for purposes of determining whether any other dependency D' obeys C.

With these two components in place, the facilitator effects in Mandarin topicalization are straightforwardly derived. (21a) illustrates how the presence of a low facilitator suffices. In this case, the first movement step crosses a low facilitator (e.g., a manner adverb), thereby satisfying the requirement of crossing in (18); the second step (from Spec, *v*P to Spec, TopP) is exempted from anti-locality evaluation, by the PMC in (20). (21b) illustrates how the presence of a high facilitator suffices. In this case, while the first movement step (from Comp, V to Spec, *v*P) violates (18), the second step crosses a high facilitator (e.g., a temporal adverb, an aspect or negation), thereby satisfying (18). Since both movement steps share the same Goal (the object), the PMC applies and the first movement step becomes licit.⁶ Note that in external topicalization, the second movement step from Spec, *v*P to Spec, TopP always crosses the subject in Spec, IP, which serves as a high facilitator.

- (21) a. *Low facilitators ameliorate Comp-to-Spec anti-locality violation*
- ... [TopP **Obj_i**] [_{Top'} Top ... [_{vP} _i [_{v'} AdvP_{manner} v+V [_{VP} t_V] _i ...]
- ✓ Spec-to-Spec by PMC ✓ crossing
- b. *High facilitator ameliorate Spec-to-Spec anti-locality violation*
- ... [TopP **Obj_i**] [_{Top'} Top ... [_{AspP/NegP} AdvP_{temp}/Asp/Neg [_{vP} _i [_{v'} v+V [_{VP} t_V] _i ...]]
- ✓ crossing ✓ Comp-to-Spec by PMC

⁵The PMC is motivated by obviation of locality constraints such as the Phase Impenetrability Condition, i.e., “phase-unlocking” (see e.g., Rackowski and Richards 2005, van Urk and Richards 2015, Branen 2018, Halpert 2019, Carstens 2023, Lee and Yip 2024; a.o.). The idea that the PMC applies to *anti*-locality, to the best of our knowledge, is only explored in Toquero-Pérez (2022:171-175), though in a different empirical domain and with a very different implementation. For Toquero-Pérez (2022), Agree is the key operation that exempts movement from anti-locality constraints, which explains why subjects may move from Spec, vP to Spec, TP. We, however, maintain the original spirit of the PMC that the same constraint only needs to be satisfied once (for a given Probe/Goal). Given that Agree is not anti-local to begin with, we contend that, in the case of Mandarin topicalization, agreement between the Top head and the topicalized phrase does not ameliorate anti-locality violations incurred on the movement path. We leave the comparison of these two implementations of the PMC to future work.

⁶This implies that the PMC should be representational and evaluated in a certain domain (e.g., a CP phase) rather than strictly derivational (i.e., by each movement step), allowing a later step to salvage an earlier one.

4.2 Against one-fell-swoop derivations

Since one facilitator suffices, an intuitive alternative is to analyze (internal) topicalization as involving *one-fell-swoop* movement, without intermediate movement to Spec, νP , as illustrated in (22a), or as involving direct adjunction to the projection in which a low or a high facilitator is hosted, as illustrated in (22b).

(22) (Untenable) *One-fell-swoop internal topicalization ...*

a. ... *without intermediate movement to Spec, νP*

[_{IP} Subj ... [_{TopP} **Obj_i** [_{TopP'} Top] ... [_{νP} $\bar{c}$ _i [_{ν'} $\nu+V$ [_{VP} t_V $\bar{c}$ _i ...

b. i. ... *as νP adjunction, when the low facilitator is present*

[_{IP} Subj ... [_{νP} **Obj_i** [_{ν'} AdvP_{manner} $\nu+V$ [_{VP} t_V $\bar{c}$ _i ...

ii. ... *as AspP/NegP adjunction, when the high facilitator is present*

[_{IP} Subj ... [_{AspP/NegP} **Obj_i** [_{Asp'/Neg'} AdvP_{temp} / Asp / Neg] [_{νP} $\nu+V$ [_{VP} t_V $\bar{c}$ _i ...

We suggest that these alternatives are untenable. First, the derivation in (22a) involves movement from Comp, V directly to Spec, TopP crossing both the complex head $\nu+V$ and Top. It does not incur a position-based or crossing-based anti-locality violation. Thus, (22a) incorrectly predicts that (internal) topicalization should not require any facilitators, contrary to fact. Second, although the derivations in (22b) do require either a low or a high facilitator, they are incompatible with the empirical generalizations established in section 2: that internal topicalization proceeds successive-cyclically via Spec, νP , and that its landing site must be structurally higher than AspP when it is present.⁷

4.3 Comparison with Newman (2020)

Newman's (2020) motivating data for the crossing-based view of anti-locality comes from facilitator effects in middles. In English, for a class of verbs like *cut* and *wash*, either a high facilitator (e.g., a modal or negation) or a low facilitator (e.g., a post-verbal adverb) is required for the construction to be well-formed.⁸

(23) *Facilitator effects in English middles* (Newman 2020:25, adapted)

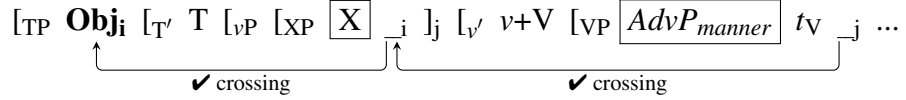
- a. The bread *(should/might/will/didn't) cut. (*high facilitator required*)
 b. The bread cut *(easily/quickly/like molasses). (*low facilitator required*)

⁷Note that an analysis assuming the possibility of νP adjunction would incorrectly predict that an internal topic can always occur below AspP, contrary to fact.

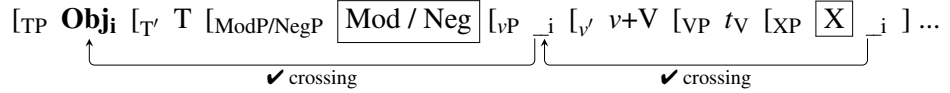
⁸Newman (2020) also discusses three other cases: (i) verbs like *read* and *wear*, which require only low facilitators; (ii) verbs like *steal* and *defeat*, which require both high and low facilitators; (iii) verbs like *know* and *own*, which cannot form middles at all. There does not seem to be verbs that require only high facilitators.

Newman (2020) proposes that middle constructions involve successive-cyclic movement of the underlying object from Comp, V to Spec, TP via Spec, vP. Thus, the facilitator effects in (23) parallel those in Mandarin topicalization, where one facilitator ameliorates anti-locality violations in two different movement steps. Newman (2020) proposes to embed the object in an XP (headed by a null X) selected by the verb. In one step, the XP moves as a whole, thus requiring a facilitator; in the other step, the object is extracted from XP, with X serving as a facilitator, as illustrated in (24). The difference between the requirement of a low vs. a high facilitator lies in whether XP is pied-piped in the first movement step.

- (24) a. *Middles: low facilitator + X as high facilitator*



- b. *Middles: high facilitator + X as low facilitator*



We argue that such an analysis cannot be extended to Mandarin topicalization, in view of cases involving more than two successive-cyclic steps. In (25), the object undergoes internal topicalization across two non-finite clausal boundaries, with no (additional) facilitator required. Under our account, this is expected. As illustrated in (26), there are three movement steps. Crucially, the intermediate step from Spec, vP3 to Spec, vP2 satisfies the requirement of crossing (by crossing the second verb *changshi* ‘try’), thereby allowing the initial and final steps to be exempted from anti-locality evaluation under the PMC.

- (25) Zhangsan_i dasuan [PRO_i **zhe-jian shi** changshi [PRO_i zuo]].
 Zhangsan plan this-CL matter try do
 Lit. ‘Zhangsan plans to, this matter, try to do (it).’

- (26)
-
- Diagram (26) illustrates three successive movement steps for the object Obj_i . Step 1: Movement from [IP1 ZS_i plan] to [IP2 PRO_i ...] is marked 'Spec-to-Spec by PMC'. Step 2: Movement from [IP2 PRO_i ...] to [IP3 PRO_i ...] is marked 'crossing' and involves crossing the verb *try*. Step 3: Movement from [IP3 PRO_i ...] to the final position is marked 'Comp-to-Spec by PMC'.

Under Newman’s (2020) proposal with $[_{XP} X Obj]$, however, X can only be counted as a facilitator *once* for a movement step. Either the initial or the final step in (26) would incur an anti-locality violation without assuming the PMC. This incorrectly predicts that an additional facilitator on the initial or final step is required, which is not the case here. To derive (25), one would have to posit further embedding of the object to create the additional facilitator, as in $[_{XP} X [_{YP} Y Obj]]$. But such further embedding cannot be assumed *uniformly*: if all objects were doubly embedded in Mandarin, then movement involving only two steps should never require a facilitator, contrary to fact.

5. Conclusion

In conclusion, we have established a novel generalization: topicalization in Mandarin requires either a high facilitator (above νP) or a low facilitator (within νP) on the movement path, and is otherwise illicit; such facilitator effects are only visible in (IP-)internal topicalization, because in (IP-)external topicalization the subject at Spec, IP serves as a (high) facilitator. The facilitator effects in Mandarin topicalization parallel those in a class of English middles, as reported by Newman (2020). At first glance, such facilitator effects are peculiar: both Mandarin topicalization and English middles involve successive-cyclic movement via $\nu P(s)$, and yet one facilitator suffices to circumvent anti-locality violations on multiple movement steps.

We have argued that the facilitator effects arise from the combination of the *crossing*-based view of anti-locality (Newman 2020), which unifies Comp-to-Spec and Spec-to-Spec anti-locality constraints, and the *Principle of Minimal Compliance* (PMC, Richards 1998), which allows the requirement of crossing to be satisfied only once for the same Goal. As a result, once a step in a successive-cyclic movement chain crosses a facilitator, the other steps in the same chain are exempted from anti-locality evaluation under the PMC.

Our proposal sheds new light on the nature of internal topicalization in Mandarin and the PMC. Specifically, we have demonstrated that anti-locality provides a principled explanation for the (un)availability of internal topicalization in non-finite clauses, based on which we contend that internal topicalization is *not* a test of finiteness, nor IP/TP clause size (*pace* Huang 1982, 1989, 2022, Li 1990, Tang 2000, Lin 2011, 2012, 2015, Grano 2012, 2015, N. Huang 2018). Furthermore, we have suggested that the PMC applies not only to standard locality constraints, such as the Phase Impenetrability Condition, but also to *anti*-locality constraints, resonating Toquero-Pérez's (2022) idea. Accordingly, the PMC should be understood as a *general* principle that conditions operations in the grammar.

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